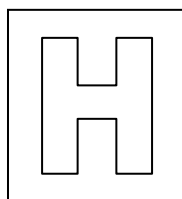


Candidate Name: _____

Class Adm No

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2011 Preliminary Examination II

Pre-university 3

H2 Biology

9648/03

19 September 2011

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This question paper consists of 16 printed pages.

[Turn over

Answer **all** questions.

1. Huntington's disease (HD) results from the inheritance of an autosomal dominant allele.

- (a) Genetic screening for HD can be carried out, using a process similar to genetic fingerprinting, to find the length of a repeated triplet, or 'stutter', in an allele.

After treatment with a restriction enzyme, fragments of DNA of different lengths are separated by gel electrophoresis.

- (i) Describe the role of a restriction enzyme in this technique.

.....

 [3]

- (ii) Explain why fragments of DNA of different lengths can be separated by gel electrophoresis.

.....

 [3]

- (b) Fig. 1.1 shows a family's history of HD.

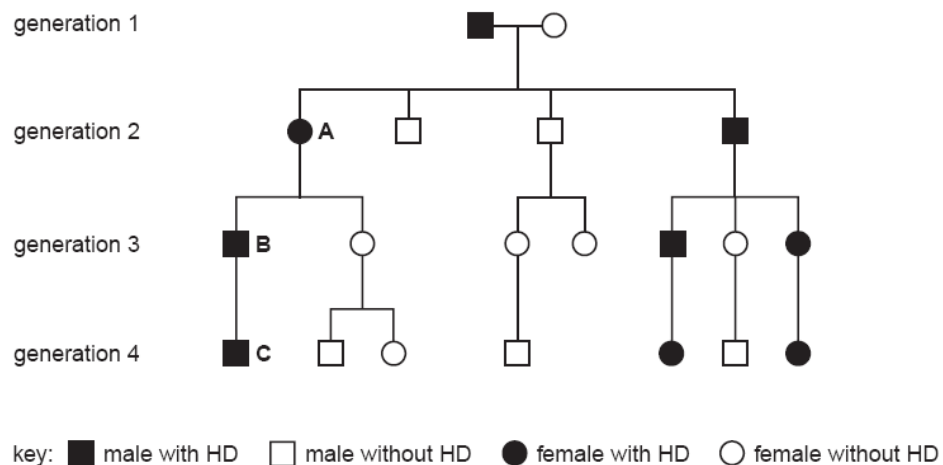


Fig. 1.1

DNA from individuals **A**, **B** and **C** from the family shown in Fig. 1.1 was analysed. The resulting banding patterns are shown in Fig. 1.2.

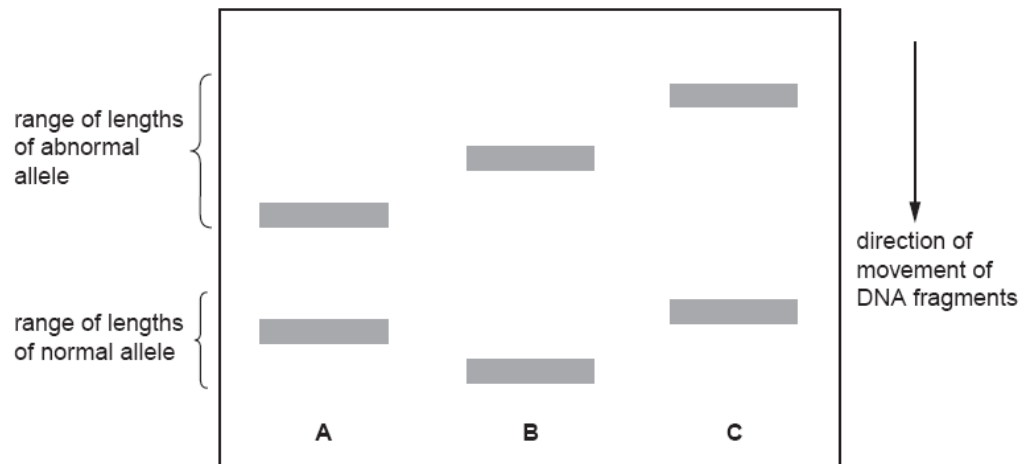


Fig. 1.2

Suggest why the DNA of the following bands in Fig. 1.2 are not the same length:

(i) the three normal alleles

.....
 [1]

(ii) the three abnormal alleles

.....
 [1]

Down's syndrome in humans usually results from acquiring an extra chromosome 21 from the mother.

A small section of chromosome 21, the so-called Down's syndrome critical region (DSCR) causes the abnormalities associated with the syndrome.

Precise regulation of gene expression by a particular group of transcription factors is essential for normal development. In inactive cells the transcription factors are found in the cytoplasm, but in active cells they move into the nucleus.

In humans, a gene, D, has been identified in the DSCR that affects a transcription factor. Mice that have been genetically engineered to over-express D show many of the symptoms of Down's syndrome.

The nuclear / cytoplasmic ratio of the transcription factors and the expression of the genes that they control was measured in the genetically engineered mice and in normal mice. The results are shown in Table 1.3.

	normal mice	mice genetically engineered to over-express D
nuclear / cytoplasmic ratio of transcription factors	2.75	0.75
relative expression of genes controlled by these transcription factors	×12	×1

Table 1.3

(c) With reference to the information given in Table 1.3 and information above,

(i) describe the effect of the over-expression of the human gene, D, in mice;

.....

 [2]

(ii) suggest why the genetically engineered mice show many of the symptoms of Down's syndrome.

.....

 [3]

- (d) Suggest what benefits and hazards arise from genetically engineering mice to express human genes.

Benefits

.....

.....

.....

Hazards

.....

.....

..... [4]

[Total: 17]

2. One alternative to the use of embryonic stem cells in human therapy are recently-developed methods of genetically reprogramming the nuclei of differentiated adult cells so that they regain the pluripotency of embryonic stem (ES) cells.

An induced pluripotent stem cell, or IPS cell, is a stem cell that has been created from an adult cell such as a skin, liver, stomach or other mature cell through the introduction of genes that reprogram the cell and transform it into a cell that has all the characteristics of an embryonic stem cell.

So far, IPS cells appear to exhibit the same key features of ES cells: the ability to differentiate from a blank-slate state to any of the 220 types of cells in the human body, and the ability to reproduce indefinitely in culture.

- (a) Stem cells can be divided into four main types: adult stem cells, embryonic stem cells, fetal stem cells and umbilical stem cells.

- (i) Identify one of the four main types of stem cell which can differentiate into a limited range of tissue?

..... [1]

- (ii) Explain how will the potency of the stem cell you have identified in (i) differs from IPS cells.

.....

 [2]

- (b) Suggest why a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell.

.....
 [1]

- (c) Besides the problem of availability of human ES cells, explain why human IPS cells are good substitute to human ES cells.

.....
 [1]

Both stem cells and cancer cells have the ability to proliferate

(d) Identify two ways which stem cells and cancer cells differ:

Stem Cells	Cancer Cells

[2]

(e) State one normal role of stem cells in living organism.

.....

..... [1]

[Total: 8]

3. Many Cowpeas, *Vigna unguiculata*, are leguminous plants which are used as forage crops. The plants contain an inhibitor of the enzyme trypsin. Trypsin is a protease that is found in the digestive tracts of many animals including insects. This provides the plant with resistance against various insect pests.

The cowpea trypsin inhibitor (CpTi) is a small polypeptide of 80 amino acids. The allele coding for its production has been successfully transferred to other crops, such as potato and tobacco, to increase their yield.

- (a) Leaf cells from tobacco plants were genetically engineered to produce CpTi by means of a plasmid from the bacterium, *Agrobacterium tumefaciens*. The plasmid included:

- the coding sequence for CpTi
- a regulatory sequence known to work in plants
- a sequence giving resistance to kanamycin, an antibiotic which kills both bacterial cells and plant cells.

- (i) Describe two properties of bacterial plasmids that allow them to be used as cloning vectors.

1

.....

2

..... [2]

- (ii) Explain the importance of the presence, in the plasmid, of the sequence giving resistance to kanamycin.

.....

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.....

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..... [3]

- (b)** The successfully genetically engineered leaf cells were grown into plantlets using tissue culture.
Outline a procedure for growing plantlets from cells in tissue culture.

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..... [5]

The effectiveness of CpTi against insect pests was tested using two groups of tobacco plants.

- Plants genetically engineered to express the allele coding for CpTi.
- Control plants which had been genetically engineered by inserting the allele coding for CpTi in the wrong orientation. These plants did not produce CpTi.

Equal numbers of each type of plant were infested with equal numbers of newly hatched larvae of an insect pest and left for seven days.

The mean percentage of leaf area eaten and the mean mass of surviving insects were then found. The results are shown in Fig. 3.1.

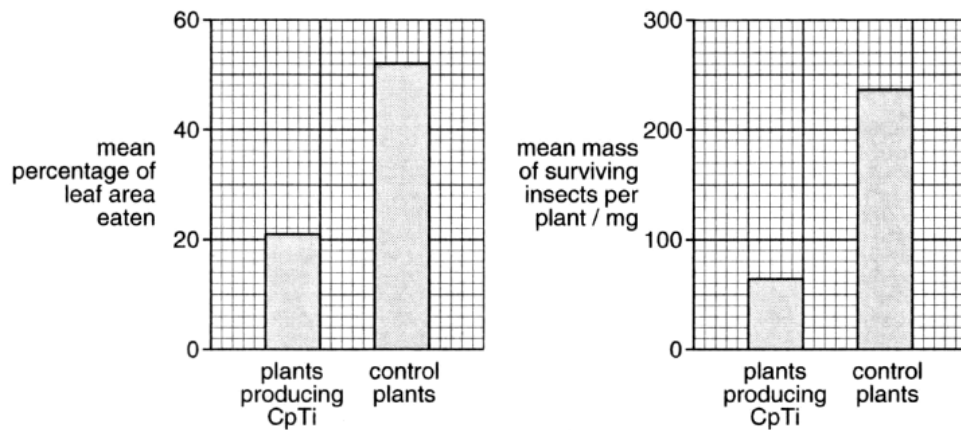


Fig. 3.1

With reference to Fig. 3.1,

(c)

(i) describe the effect of CpTi on yield of tobacco leaves

.....

 [2]

(ii) suggest an explanation of the effect of the trypsin inhibitor, CpTi, on insects.

.....

 [3]

[Total: 15]

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4. Transformation of plasmid DNA into *E. coli* using the heat shock method is a basic technique of molecular biology. It consists of inserting a foreign plasmid or ligation product into bacteria.

After a short incubation in ice, a mixture of chemically competent bacteria and DNA is placed at a warm temperature for 45 seconds (heat shock) and then placed back in ice. This traditional protocol can be used successfully to transform most commercially available competent bacteria. Fig. 4.1 shows a simple illustration of bacteria transformation through heat shock.

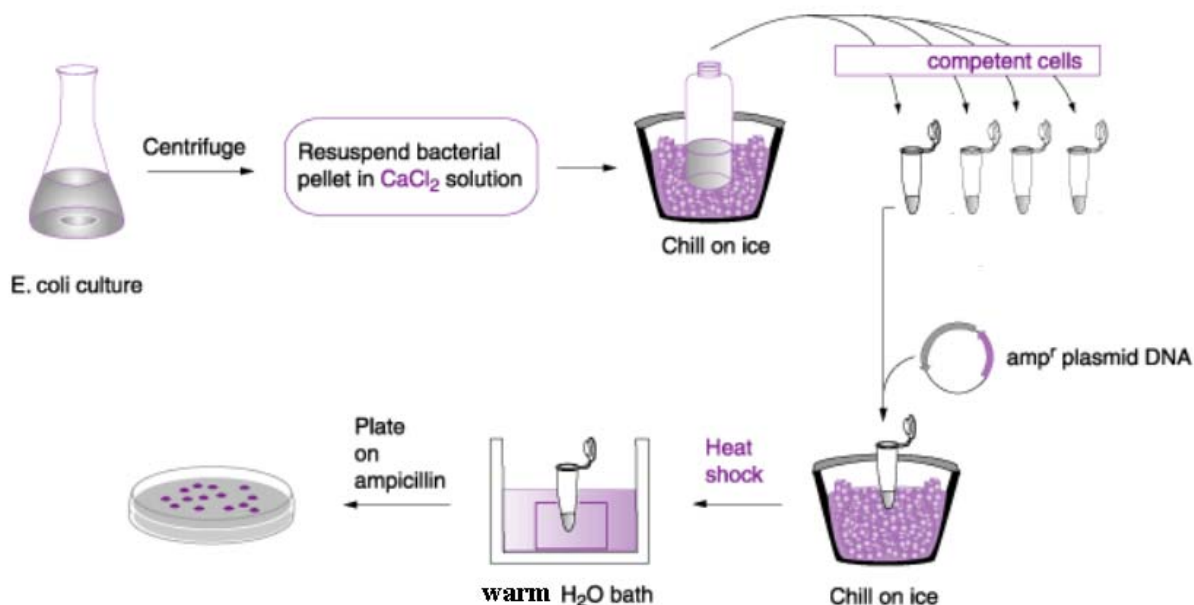


Fig. 4.1

You are to design an experiment to investigate the effect of varying the heat-shock temperature on the transformation of the bacterium *Escherichia coli* using a plasmid vector.

[Total: 12]

- Your plan should have a clear and helpful structure to include:
- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

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[illegible]

[illegible]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5. With reference to genetic diseases,

(a) Describe SCID (severe combined immunodeficiency) [8]

(b) Explain gene therapy treatment, using both viral and non-viral delivery systems. [6]

(a) Discuss the factors which prevent gene therapy from becoming an effective treatment. [6]

[Total: 20]